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Electro-fluidic remote drive

Abstract

A drive system includes: a remote drive configured for being driven by an electrical motor; and an actuator spaced apart from and fluidically coupled with the remote drive and configured for being fluidically powered by the remote drive.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This is a non-provisional application based upon U.S. provisional patent application Ser. No. 63/419,934, entitled “ELECTRO-FLUIDIC REMOTE DRIVE”, filed Oct. 27, 2022, which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to robotics, and, more particularly, to systems for moving movable members of a robot.

2. Description of the Related Art

(3) Actuators are devices characterized by one or more translating or rotating members (generally, movable members) which are moved relative to a stationary member by motive ways such as an electric motor or fluid driven piston. It is often desirable to minimize the mass of the actuator, especially in applications involving integration of the actuator as an end effector onto robots, where the mass of the actuator correspondingly reduces the mass of the workpiece that the robot can

manipulate. Physically separating the components of the actuator responsible for moving and guiding the movable member from those components responsible for generating the motive force provides an effective way of reducing the mass that the robot must manipulate. Historically, in the prior art, this component separation has been accomplished by the use of fluid power transmission.

(4) A pump or compressor, often located some distance away from the actuator, provides pressurized fluid through an appropriate network of valves, pipes, and flexible tubes to a piston that moves the movable member. Systems using such a centralized pump or compressor suffer from leaks along the length of the pipes and tubes used to convey the pressurized fluid from the pump or compressor to the actuator. Further, centralized electrically operated pumps or compressors also typically exhibit poor efficiency in converting electrical energy into compressed fluid. Further, such pumps and compressors often produce objectionable noise and heat that must be addressed. The loss of compressed fluid due to leaks, generally poor energy efficiency, and objectionable noise and heat of centralized pumps and compressors often necessitate the use of an electrical motor to directly convert electric current into motive force with the motor in close physical proximity to the actuator in order to achieve a desired efficiency of electric to translational or rotational motion conversion. Such configurations suffer the disadvantage of adding the relatively large mass of the motor to the total mass that a robot or machine must move when the actuator is used as an end effector.

(5) What is needed in the art is a system for moving movable members with improved efficiency that at least partially overcomes the aforementioned disadvantages.

SUMMARY OF THE INVENTION

(6) The present invention provides a drive system including a drive and an actuator, the drive being remotely located relative to the actuator.

(7) The invention in one form is directed to a drive system which includes: a remote drive configured for being driven by an electrical motor; and an actuator spaced apart from and fluidically coupled with the remote drive and configured for being fluidically powered by the remote drive.

(8) The invention in another form is directed to a drive of a drive system, the drive including: the drive, which is a remote drive that is configured for being driven by an electrical motor, is configured for being spaced apart and fluidically coupled with an actuator of the drive system, and is configured for fluidically powering the actuator.

(9) The invention in yet another form is directed to a method of using a drive system, the method including the steps of: providing that the drive system includes a remote drive and an actuator, the remote drive being configured for being driven by an electrical motor, the actuator being spaced apart from and fluidically coupled with the remote drive; and fluidically powering the actuator by the remote drive.

(10) An advantage of the present invention is that it provides an improved device to electrically drive a fluid powered actuator. Such an electro-fluidic drive is a remote drive (rather than a centralized drive, such as a centralized pump or compressor) that traverses the limitations discussed for the prior art by incorporating a way to physically separate components responsible for generating a motive fluid from the components responsible for generating the motion of a movable actuator member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above-mentioned and other features and advantages of this invention, and the manner of attaining them, will become more apparent and the invention will be better understood by reference to the following description of embodiments of the invention taken in conjunction with the

accompanying drawings, wherein:
(2) FIG. **1** shows a schematic of an embodiment of drive system including a remote drive and an actuator, in accordance with an exemplary embodiment of the present invention;
(3) FIG. **2** shows an isometric view of another embodiment of a remote drive which uses air as the working fluid, in accordance with an exemplary embodiment of the present invention;
(4) FIGS. **3**, **4**, and **5** show partially exploded isometric views of the remote drive of FIG. **2**;
(5) FIG. **6** shows a top view of the remote drive of FIG. **2**;
(6) FIG. **7** shows a cross-sectional view of the remote drive taken along line **7-7** in FIG. **6**;
(7) FIG. **8** shows an exploded view of piston assemblies **114**/**115** shown in FIGS. **5** and **7**;
(8) FIG. **9** shows top view of piston assembly **114** of the remote drive of FIG. **2**;
(9) FIG. **10** shows a cross-sectional view of piston assembly **114** taken along line **10-10** in FIG. **9**;
(10) FIG. **11** shows top view of piston assembly **115** of the remote drive of FIG. **2**;
(11) FIG. **12** shows a cross-sectional view of piston assembly **115** taken along line **12-12** in FIG. **11**;
(12) FIG. **13** shows an isometric view of yet another embodiment of a remote drive which uses air as the working fluid, in accordance with an exemplary embodiment of the present invention;
(13) FIG. **14** shows a partially exploded isometric view of the remote drive of FIG. **13**;
(14) FIG. **15** shows an exploded isometric view of cylinder assembly **300** of FIG. **14**;
(15) FIG. **16** shows an exploded isometric view of cylinder assembly **400** of FIG. **14**;
(16) FIG. **17** shows an exploded view of piston assemblies **313**/**413** shown in FIGS. **15** and **16**;
(17) FIG. **18** shows an isometric view of a piston movement system of the remote drive of FIG. **13**;
(18) FIG. **19** shows a schematic of another embodiment of a drive system, which includes the remote drive of FIG. **13** and which is configured to provide a source of pressurized air to an external pneumatic circuit, in accordance with an exemplary embodiment of the present invention;
(19) FIG. **20** shows a schematic of yet another embodiment of a drive system, which includes the remote drive of FIG. **13** and which is configured to provide a source of rarified air (i.e. a partial vacuum) to an external pneumatic circuit, in accordance with an exemplary embodiment of the present invention;
(20) FIG. **21** shows another embodiment of a drive system including a pneumatically powered, cable driven rotary actuator based on the cylinder assembly of FIG. **16**, in accordance with an exemplary embodiment of the present invention;
(21) FIG. **22** shows an exploded isometric view of the cable driven rotary actuator of FIG. **21**;
(22) FIG. **23** shows an exploded view of the piston assembly **713** shown in FIG. **22**;
(23) FIG. **24** shows an isometric view of yet another embodiment of a remote drive of a drive system, the remote drive using air as the working fluid and being in part similar to the remote drive of FIG. **13**, in accordance with an exemplary embodiment of the present invention;
(24) FIG. **25** shows a partially exploded isometric view of the remote drive of FIG. **24**;
(25) FIG. **26** shows an exploded isometric view of end cap **804** and cable guide assemblies **850**/**851** of FIG. **25**;
(26) FIG. **27** shows an exploded isometric view of cable guide assembly **850** of FIG. **26**;
(27) FIG. **28** shows an isometric view of a piston movement system of the remote drive of FIG. **24**; and
(28) FIG. **29** illustrates a flow diagram showing a method of using a drive system, in accordance with an exemplary embodiment of the present invention.
(29) Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate embodiments of the invention, and such exemplifications are not to be construed as limiting the scope of the invention in any manner.

DETAILED DESCRIPTION OF THE INVENTION

(30) Referring to FIG. **1**, there is shown a drive system **18**, which includes each of the structures shown in FIG. **1**. Drive system **18** includes an electrically powered motor **14**, a remote drive **1**

(remote drive **1** can be said to include motor **14**), an actuator **2**, and tubing **3**A, **3**B. Remote drive **1** is fluidically coupled to fluid-powered actuator **2**, wherein remote drive **1** is configured for being driven by motor **14**, and actuator **2** is spaced apart from and fluidically coupled with remote drive **1** and is configured for being fluidically powered by remote drive **1**. Remote drive **1** additionally includes toothed racks **11**A, **11**B, pinion **12**, cylinders **6**A, **6**B, and pistons **5**A, **5**B. Further, remote drive **1** includes a rack and pinion system **19** (which includes racks **11**A, **11**B and pinion **12**) and piston assemblies **20**A, **20**B (which respectively includes piston **5**A, cylinder **6**A and piston **5**A, cylinder **6**B) which are coupled with rack and pinion system **19**. Actuator **2** includes cylinder **9**, piston **8**, and rod **10**. FIG. **1** shows that remote drive **1** is connected to a fluidically powered actuator **2** with tubing **3**A and **3**B so as to allow the actuator **2** to be physically separated from remote drive **1**. Tubing **3**A connects volume **4**A formed by movable piston **5**A and stationary cylinder **6**A within remote drive **1** to volume **7**A formed by movable piston **8** and stationary cylinder **9** of actuator **2**. In an analogous manner, tubing **3**B connects volume **4**B formed by movable piston **5**B and stationary cylinder **6**B within remote drive **1** to volume **7**B formed by movable piston **8** and stationary cylinder **9** of actuator **2**. Rod **10**, attached to piston **8**, communicates (or, commutates) the translation of piston **8** within cylinder **9** to any device mechanically connected to actuator **2**. Toothed racks **11**A and **11**B are mechanically attached to movable pistons **5**A and **5**B, respectively. Toothed pinion **12** engages racks **11**A and **11**B such that rotation of the pinion **12** causes a proportional translation of the racks **11**A, **11**B. The output shaft **13** of electrically powered motor **14** is mechanically coupled to pinion **12** so that the motor **14** can selectively rotate the pinion **12**. Working fluid **15**A fills the volume including remote drive volume **4**A, the volume within connecting tubing **3**A, and actuator volume **7**A. In an analogous manner, working fluid **15**B fills the volume including remote drive volume **4**B, the volume within connecting tubing **3**B, and actuator volume **7**B. These connections between volumes **4**A and **4**B within remote drive **1** to corresponding volumes **7**A and **7**B, respectively, within actuator **2** create a fluidic way of communication between remote drive **1** and actuator **2**. As motor **14** is activated to rotate pinion **12** in a counterclockwise (CCW) direction, as shown by arrow **16**, rack **11**A is driven to the left (in the direction of arrow **14**A) by the engagement of pinion **12**, simultaneously moving piston **5**A to the left to reduce volume **4**A, thereby forcing a quantity of working fluid **15**A from remote drive **1** into actuator **2**. The CCW rotation of pinion **12** causes a simultaneous translation rack **11**B to the right, in the direction of arrow **14**B, moving piston **5**B to the right to increase volume **4**B, thereby extracting a quantity of working fluid **15**B from actuator **2** into remote drive **1**. The action of working fluid **15**A entering actuator volume **7**A and working fluid **15**B exhausting from actuator volume **7**B exerts forces on actuator piston **8**, this action acting to move piston **8** and attached rod **10** downward, in the direction shown by arrow **17**. It will be apparent to one skilled in the art that rotation of pinion **12** by motor **14** in a clockwise (CW) direction will force working fluid **15**B from remote drive volume **4**B into actuator volume **7**B while working fluid **15**A from actuator volume **7**A is exhausted into remote drive volume **4**A, thereby moving actuator piston **8** and rod **10** upwards, opposite to the direction shown by arrow **17**. In this manner, the rotation of motor shaft **13** is proportionally coupled to the translation of actuator rod **10**. It will also be apparent that the working fluid **15**A and **15**B can be either a compressible gas, such as air, to create a pneumatically coupled system, or an incompressible liquid, such as water or oil, to create a hydraulically coupled system.

(31) Referring now to FIGS. **2-12**, there is shown another embodiment the drive system according to the present invention, namely, drive system **126**, drive system **126** including another embodiment of the remote drive according to the present invention, namely, remote drive **100**, and an actuator **127**. Remote drive **100** includes motor **101**, fasteners **102**, body **103**, jaw **104**, spider **105**, coupling **106**, radial bearing **107**, pinion **108**, radial ball bearing **109**, retaining ring **110**, bearing washers **111**A, **111**B, threaded plugs **112**A, **112**B, racks **113**A, **113**B, piston assemblies **114**, **115**, cylinder tubes **116**A, **116**B, end cap **117**, threaded ports **119**A, **119**B, tierods **120**, **124**, tierod

nuts **121**, **125**, rack cover **122**, and end cap **123**. Piston assemblies **114**, **115** include, respectively, pins **114**A, **115**A, inner pistons **114**B, **115**B, slots **114**C, **115**C, outer pistons **114**D, **115**D, O-rings **114**E, **115**E, and seals **114**F, **115**F. Further, remote drive **100** includes a rack and pinion system **128** that includes racks **113**A, **113**B and pinion **108**, and piston assemblies **114**, **115** are coupled with rack and pinion system **128**. Inner pistons **114**B, **115**B and outer pistons **114**D, **115**D are configured for sliding relative to one another and thereby for selectively opening and closing a working medium passage **129** therebetween. In general, similar to drive system **18**, motor **101** of remote drive **100** moves racks **113**A, **113**B via pinion **108**, racks **113**A, **113**B moving piston assemblies **114**, **115**, causing a working fluid such as air to flow in or out of ports **119**A, **119**B, so as to move a working device thereby, such as an actuator similar to actuator **2**. Thus, remote drive **100** uses air as a working fluid and incorporates rack and pinion system **128** to convert rotational motion from electric motor **101** into linear motion of pistons assemblies **114**, **115** pneumatically coupled to a pneumatically driven actuator. This embodiment of the remote drive includes a provision (discussed below) to replenish the working fluid within the pneumatic circuit formed between the remote drive **100** and the driven actuator, some portion of which (the working fluid) may be lost through leakage each time the actuator is cycled. Remote drive **100** includes motor **101**, which can be an alternating current or direct current motor, servomotor, gearmotor or servo-gearmotor, that is attached with mechanical fasteners **102** to body **103**. The output shaft of motor **101** is rotationally coupled to pinion **108**, through a commercially available shaft coupling, such as the Rotex series produced by KTR Corporation, which includes jaw **104**, spider **105**, and coupling **106**. Pinion **108** is supported by radial ball bearings **107** and **109**. Although radial ball bearings **107**, **109** are shown, it will be understood by one skilled in the art that sintered metal or polymer bushings can be similarly used to support pinion **108**. Retaining ring **110** retains bearing **109** within body **103**. The teeth of pinion **108** engage complementary teeth in upper rack **113**A and lower rack **113**B, such that rotation of pinion **108** is converted into proportional translation of racks **113**A, **113**B. Bearing washers **111**A and **111**B support racks **113**A and **113**B, respectively, within body **103**. Bearing washers **111**A and **111**B can be fabricated from suitable oil-filled sintered metal or polymer, such as the thrust bearing washers manufactured by Igus Corporation. Threaded plugs **112**A and **112**B bear against washers **111**A and **111**B, respectively, which are contained within complementary threaded bores in body **103**. In this manner, washers **111**A and **111**B are constrained from radial movement by contact with the threaded bores in body **103**, but are free to translate, under the action of plugs **112**A and **112**B so that racks **113**A and **113**B can be positioned about pinion **108** so as to align the pitch line of racks **113**A, **113**B with the pitch line of pinion **108**. Pins **114**A and **115**A pass though complementary holes in racks **113**A and **113**B and complementary holes in inner pistons **114**B and **115**B to attach piston assemblies **114** and **115** to racks **113**A and **113**B, respectively. The ends of pins **114**A and **115**A engage slots **114**C and **115**C in outer piston **114**D and **115**D, respectively (see FIGS. **7** and **8**), such that inner piston **114**B/**115** B is free to translate relative to outer piston **114**D/**115**D within the extents of slots **114**C/**115**C. O-ring **114**E/**115**E is disposed in a complementary groove in inner piston **114**B/**115**B to seal the periphery of inner piston **114**B, **115**B against the interior of outer piston **114**D/**115**D when pin **114**A/**115**A has translated to the extent of slot **114**C/**115**C in the direction of seal **114**F/**115**F. Seal **114**F seals the periphery of outer piston **114**D against the inner diameter of upper cylinder tube **116**A. Seal **115**F seals the periphery of outer piston **115**D against the inner diameter of lower cylinder tube **116**B. End cap **117** encloses and seals the ends of cylinder tubes **116**A and **116**B. The combination of piston assembly **114**, tube **116**A, and cap **117** form volume **118**A, analogous to volume **4**A formed in the remote drive of FIG. **1**. The combination of piston assembly **115**, tube **116**B, and cap **117** form volume **118**B, analogous to volume **4**B formed in the remote drive of FIG. **1**. Upper threaded port **119**A in cap **117** allows for the establishment of a pneumatic connection between volume **118**A and an external pneumatically driven actuator analogous to the connection between volumes **4**A and **7**A in the remote drive of FIG. **1**. Lower threaded port **119**B in cap **117** allows for the establishment of a pneumatic connection between volume **118**B and an external pneumatically driven actuator analogous to the connection between volumes **4**B and **7**B in the remote drive of FIG. **1**. One end of threaded tierods **120** thread into complementary holes in body **103**. The opposing end of threaded tierods **120** pass through complementary holes in end cap **117** and are threaded into to tierod nuts **121** to retain cap **117** onto tubes **116**A and **116**B. Rack cover **122** abuts body **103** to protect the portion of racks **113**A and **133**B which protrude to the left of body **103**. End cap **123** covers the opposing end of rack cover **122**. One end of threaded tierods **124** thread into complementary holes in body **103**. The opposing end of threaded tierods **124** pass through complementary holes in end cap **123** and are threaded into tierod nuts **125** to retain cap **123** onto cover **122**.

(32) Referring now to FIGS. **9** and **10**, movement of piston assembly **114**, under the action of upper rack **113**A moving in the direction of arrow **114**G, causes a simultaneous translation of inner piston **114**B, which is attached to rack **113**A by pin **114**A. Friction forces **114**J, acting between compressed seal **114**F and the inner diameter wall of tube **116**A (see also FIG. **7**), resist the translation of outer piston **114**D, which results in a relative displacement of inner piston **114**B with respect to outer piston **114**D until pin **114**A translates sufficiently to contact the left-most end of slot **114**C. This relative displacement causes O-ring **114**E to leave the confines of the smaller of the two concentric bores that pass through the entirety of outer piston **114**D. The removal of O-ring seal **114**E from the bore in outer piston **114**D creates an annular passage around the seal **114**E through which external air can flow, as indicated by arrows **114**H, between the outer diameter of inner piston **114**B and the inner diameter of outer piston **114**D. In this manner, leftward translation of piston assembly **114** opens the annular passage between inner piston **114**B and outer piston **114**D, allowing external ambient air to pass into volume **118**A to replenish any air that may have leaked from the pneumatic circuit formed by the pneumatic connection between volume **118**A and an externally connected pneumatically driven actuator (not shown).

(33) Referring now to FIGS. **11** and **12**, the movement of piston assembly **115** is shown. Translation of piston assembly **115**, under the action of lower rack **113**B moving in the direction of arrow **115**G, causes a simultaneous translation of inner piston **115**B, which is attached to rack **113**B by pin **115**A. Friction forces **115**J, acting between compressed seal **115**F and the inner diameter wall of tube **116**B (see also FIG. **7**) resist the translation of outer piston **115**D, which results in a relative displacement of inner piston **115**B with respect to outer piston **114**D until pin **114**A translates sufficiently to contact the right-most end of slot **114**C. This relative displacement causes O-ring **115**E to enter the confines of the smaller of the two concentric bores that pass through the entirety of outer piston **115**D, establishing the seal between inner and outer pistons **115**B, **115**D, to close the annular air passage through piston assembly **115**. In this manner, rightward translation of piston assembly **115** closes the annular passage between inner piston **115**B and outer piston **115**D, preventing the exit of air within volume **118**B to the external environment.

(34) It will be apparent to one skilled in the art, that subsequent motion of piston assembly **114** in the direction opposite of arrow **114**G will cause inner piston **114**B to be displaced to the right relative to outer piston **114**D so that O-ring **114**E is caused to reenter the confines of the smaller bore through the outer piston **114**D, reestablishing the seal between inner and outer pistons **114**B, **114**D, to close the annular air passage through piston assembly **114**. Translation of piston assembly **114** in the direction opposite of arrow **114**G will result in simultaneous translation of piston assembly **115** in the direction opposite of arrow **115**G, causing a relative displacement of inner piston **115**B with respect to outer piston **115**D, removing O-ring seal **115**E from the bore in outer piston **115**D to create an annular passage around the seal **115**E through which external air can flow, allowing external ambient air to pass into volume **118**B to replenish any air that may have leaked from the pneumatic circuit formed by the pneumatic connection between volume **118**B and an externally connected pneumatically driven actuator (not shown). In this manner, ambient external air is allowed to pass through piston assemblies **114** and **115**, each time assemblies **114**, **115** are translating in a direction which acts to expand the volumes **118**A and **118**B, respectively.

(35) Referring now to FIGS. **13-18**, there is shown another embodiment the drive system according to the present invention, namely, drive system **213**, drive system **213** including another embodiment of the remote drive according to the present invention, namely, remote drive **200**, and an actuator **214**. Remote drive **200** includes motor **201**, fasteners **202**, **208**, **209**, **210**, motor spacer **203**, covers **204**, **207**, threaded ports **211**A, **211**B, **212**A, **212**B, pulleys **205**, **206**, cylinder assemblies **300**, **400**. Cylinder assemblies **300**, **400** include, respectively, pulleys **301**, **401**, drive shafts **302**, **402**, radial bearings **303**, **403**, retaining rings **304**, **404**, blocks **305**, **405**, radial seals **306**, **406**, cable drums **307**, **407**, cables **308**, **408**, cable pulleys **309**, **409**, pivot pins **310**, **410**, seals **311**, **411**, cylinder tubes **312**, **412**, piston assemblies **313**, **413**, anchors **314**, **414**, seals **315**, **415**, retaining rings **316**, **416**, radial bearings **317**, **417**, tierods **318**, **418**, O-ring seals **319**, **419**, end blocks **320**, **420**, radial seals **321**, **421**, cable drums **322**, **422**, cables **323**, **423**, pulleys **324**, **424**, pivot pins **325**, **425**, anchors **326**, **426**, and seals **327**, **427**. Piston assemblies **313**, **413**, respectively, include cable pulleys **313**A, **413**A, inner pistons **313**B, **413**B, pivot pins **313**C, **413**C, cable pulleys **313**D, **413**D, pivot pins **313**E, **413**E, O-rings **313**F, **413**F, slots **313**G, **413**G, outer pistons **313**H, **413**H, seals **313**J, **413**J. Further, remote drive **200** includes a cable and pulley system **215**, piston assemblies **313**, **413** being coupled with cable and pulley system **215**. Cable and pulley system **215** includes cables **308**, **408**, **323**, **423** and pulleys **205**, **206**, **301**, **401**, **309**, **409**, **324**, **424**, **313**A, **413**A, **313**D, **413**D (structure overlapping between cable and pulley system **215** and piston assemblies **313**, **413** (i.e., pulleys **313**A, **413**A, **313**D, **413**D) can be regarded as being a part of cable and pulley system **215** or piston assemblies **313**, **413**). Inner pistons **313**B, **413**B and outer pistons **313**H, **413**H are configured for sliding relative to one another and thereby for selectively opening and closing a working medium passage **129** therebetween (see FIGS. **9-12**, for, the structure, function, and operation of working medium passage **129** of piston assemblies **313**, **413** are substantially similar to those of working medium passage **129** of piston assemblies **114**, **115**). In general, similar to drive system **18**, motor **201** of remote drive **200** moves belt **206**, which rotates drive shafts **302**, **402** and drums **307**/**407**, **322**/**422**, causing spans of cables **308**/**408** and **323**/**423** to lengthen or shorten, which causes piston assemblies **313**, **413** to translate within respective tubes **312**, **412**, which causes air to flow in or out of ports **211**A/**211**B, **212**A/**212**B, so as to move a working device thereby, such as an actuator similar to actuator **2**. Thus, remote drive **200** replaces the rack and pinion system of remote drives **1** and **100** with a cable and pulley system to convert the rotational motion from the electric motor into linear motion of piston assemblies **313**, **413** fluidically coupled to a pneumatically driven actuator. Remote drive **200** includes motor **201**, which can be an alternating current or direct current motor, servomotor, gearmotor or servo-gearmotor, that is attached with mechanical fasteners **202** to motor spacer **203**. Timing belt pulley **205** is secured to the output shaft of motor **201** with setscrews (not shown). Timing belt **206** passes around timing belt pulley **205** and also around timing belt pulleys **301** and **401** so that rotation of the output shaft of motor **201** causes a simultaneous rotation of timing belt pulleys **301** and **401**. Motor spacer **203** is attached to cover **204** with threaded fasteners **208**. Cylinder assemblies **300** and **400** are attached on one end to cover **204** with threaded fasteners **209**. The opposing ends of cylinder assemblies **300** and **400** are attached to cover **207** with threaded fasteners **210**.

(36) Referring now to FIGS. **15-17**, timing belt pulley **301** is secured to drive shaft **302** with setscrews (not shown). One end of drive shaft **302** passes though radial bearing **303**, with the bearing **303** retained by retaining ring **304** within a complementary bore in end block **305**. The end of shaft **302** then passes through radial seal **306**, disposed in a complementary bore in block **305** so as to prevent air from passing between shaft **302** and block **305**, and into cable drum **307**, with setscrews (not shown) securing the drum **307** onto the end of shaft **302**. One end of cable **308** passes through a hole in drum **307**, with a plurality of turns of cable **308** wound around the periphery of drum **307**. Cable **308** may take the form of steel wire rope or be constructed from suitable polymer threads woven or braided together. Cable **308** is directed from the windings about drum **307** to wrap around the outer diameter of cable pulley **309**, which is retained in a

complementary slot in end block **305** by, and is free to pivot about, pivot pin **310** which is disposed in a complementary hole in the protruding boss of block **305**. Cable **308** passes through block **305**, O-ring seal **311** and cylinder tube **312** to engage cable pulley **313**A, which is retained in a complementary slot in inner piston **313**B by, and is free to pivot about, pivot pin **313**C which is disposed in a complementary hole in inner piston **313**B. O-ring seal **311** seals the end of tube **312** against the face of end block **305** to prevent the egress of compressed air from the end of tube **312**. Cable **308** continues around the periphery of pulley **313**A and returns through tube **312**, seal **311**, and block **305** to terminate with a plurality of turns of cable **308** wound around the periphery of capstan anchor **314** before passing through a complementary hole in anchor **314**. Anchor **314** is disposed in a complementary groove in block **305** to prevent both rotation and translation of anchor **314**. The turns of cable **308** about the body of anchor **314** exploit the so-called capstan effect, wherein any tension applied to cable **308** is progressively reduced by friction between the surface of cable **308** and the surface of anchor **314** as additional turns of cable **308** are added. This reduces the tension applied to cable **308** by the action of pulley **313**A acting upon cable **308** to a level sufficient to retain the end of cable **308** within the hole in the anchor **314** through which cable **308** passes. In an analogous manner, the turns of cable **308** wound about cable drum **307** also exploits the capstan effect to retain the opposing end of cable **308** within the complementary cable **308** hole in drum **307**. Seal **315** seals against the face of end cover **204** (see FIG. **14**) to prevent the egress of compressed air from block **305** through other than threaded port **211**A (see FIG. **14**).

(37) The opposing end of drive shaft **302** passes though radial bearing **317**, with the bearing retained by retaining ring **316** within a complementary bore in end block **320**. The end of shaft **302** then passes through radial seal **321**, disposed in a complementary bore in block **320** so as to prevent air from passing between shaft **302** and block **320**, and into cable drum **322**, with setscrews (not shown) securing drum **322** onto the end of shaft **302**. One end of cable **323** passes through a hole in drum **322**, with a plurality of turns of cable **323** wound around the periphery of drum **322**. Cable **323** may take the form of steel wire rope or be constructed from suitable polymer threads woven or braided together. Cable **323** is directed from the windings about drum **322** to wrap around the outer diameter of cable pulley **324**, which is retained in a complementary slot in end block **320** by, and is free to pivot about, pivot pin **325** which is disposed in a complementary hole in the protruding boss of block **320**. Cable **323** passes through block **320**, O-ring seal **319** and cylinder tube **312** to engage cable pulley **313**D, which is retained in a complementary slot in inner piston **313**B by, and is free to pivot about, pivot pin **313**E which is disposed in a complementary hole in inner piston **313**B. The ends of pin **313**C engage slot **313**G in outer piston **313**H, such that inner piston **313**B is free to translate relative to outer piston **313**H within the extents of slot **313**G. O-ring **313**F is disposed in a complementary groove in inner piston **313**B to seal the periphery of inner piston **313**B against the interior of outer piston **313**H when the pin **313**C has translated to the extent of slot **313**G in the direction of seal **313**F. Seal **313**F seals the periphery of outer piston **313**H against the inner diameter of cylinder tube **312**. O-ring seal **319** seals the opposing end of tube **312** against the face of end block **320** to prevent the egress of compressed air from the end of the tube. Cable **323** continues around the periphery of pulley **313**D and returns through tube **312**, seal **319**, and block **320** to terminate with a plurality of turns of cable **323** wound around the periphery of capstan anchor **326** before passing through a complementary hole in anchor **326**. Anchor **326** is disposed in a complementary groove in block **320** to prevent both rotation and translation of anchor **326**. The turns of cable **323** about anchor **326** and drum **322** exploit the capstan effect to allow the complementary cable holes in the anchor and drum to retain the respective ends of cable **323**. Seal **327** seals against the face of end cover **207** (FIG. **14**) to prevent the ingress of external ambient air into block **320** through other than threaded port **212**A (FIG. **14**). Tierods **318** surround tube **312** to mechanically join together end blocks **305** and **320**.

(38) Referring now to FIGS. **16** and **17**, timing belt pulley **401** is secured to drive shaft **402** with setscrews (not shown). One end of drive shaft **402** passes though radial bearing **403**, with the

bearing retained by retaining ring **404** within a complementary bore in end block **405**. The end of shaft **402** then passes through radial seal **406**, disposed in a complementary bore in block **405** so as to prevent air from passing between the shaft and block, and into cable drum **407**, with setscrews (not shown) securing drum **407** onto the end of shaft **402**. One end of cable **408** passes through a hole in drum **407**, with a plurality of turns of cable **408** wound around the periphery of drum **407**. Cable **408** may take the form of steel wire rope or be constructed from suitable polymer threads woven or braided together. Cable **408** is directed from the windings about drum **407** to wrap around the outer diameter of cable pulley **409**, which is retained in a complementary slot in end block **405** by, and is free to pivot about, pivot pin **410** which is disposed in a complementary hole in the protruding boss of block **405**. Cable **408** passes through block **405**, O-ring seal **411** and cylinder tube **412** to engage cable pulley **413**A, which is retained in a complementary slot in inner piston **413**B by, and is free to pivot about, pivot pin **413**C which is disposed in a complementary hole in inner piston **413**B. The ends of pin **413**C engage slot **413**G in outer piston **413**H, such that inner piston **413**B is free to translate relative to outer piston **413**H within the extents of slot **413**G. O-ring **413**F is disposed in a complementary groove in inner piston **413**B to seal the periphery of inner piston **413**B against the interior of outer piston **413**H when the pin **413**C has translated to the extent of slot **413**G in the direction of seal **413**F. O-ring seal **411** seals the end of tube **412** against the face of end block **405** to prevent the egress of compressed air from the end of tube **412**. Cable **408** continues around the periphery of pulley **413**A and returns through tube **412**, seal **411**, and block **405** to terminate with a plurality of turns of cable **408** wound around the periphery of capstan anchor **414** before passing through a complementary hole in anchor **414**. Anchor **414** is disposed in a complementary groove in block **405** to prevent both rotation and translation of anchor **414**. The turns of cable **408** about the body of anchor **414** exploit the so-called capstan effect, wherein any tension applied to cable **408** is progressively reduced by friction between the surface of cable **408** and the surface of anchor **414** as additional turns of cable **408** are added. This reduces the tension applied to cable **408** by the action of pulley **413**A acting upon cable **408** to a level sufficient to retain the end of cable **408** within the hole in anchor **414** through which cable **408** passes. In an analogous manner, the turns of cable **408** wound about cable drum **407** also exploit the capstan effect to retain the opposing end of cable **408** within the complementary cable hole in drum **407**. Seal **415** seals against the face of end cover **204** (FIG. **14**) to prevent the egress of compressed air from the block **405** through other than threaded port **211**B (FIG. **14**).

(39) The opposing end of drive shaft **402** passes though radial bearing **417**, with the bearing retained by retaining ring **416** within a complementary bore in end block **420**. The end of shaft **402** then passes through radial seal **421**, disposed in a complementary bore in block **420** so as to prevent air from passing between the shaft and block, and into cable drum **422**, with setscrews (not shown) securing drum **422** onto the end of shaft **402**. One end of cable **423** passes through a hole in drum **422**, with a plurality of turns of cable **423** wound around the periphery of drum **422**. Cable **423** may take the form of steel wire rope or be constructed from suitable polymer threads woven or braided together. Cable **423** is directed from the windings about drum **422** to wrap around the outer diameter of cable pulley **424**, which is retained in a complementary slot in end block **420** by, and is free to pivot about, pivot pin **425** which is disposed in a complementary hole in the protruding boss of block **420**. Cable **423** passes through block **420**, O-ring seal **419** and cylinder tube **412** to engage cable pulley **413**D, which is retained in a complementary slot in inner piston **413**B by, and is free to pivot about, pivot pin **413**E which is disposed in a complementary hole in inner piston **413**B. O-ring seal **419** seals the opposing end of tube **412** against the face of end block **420** to prevent the egress of compressed air from the end of the tube. Cable **423** continues around the periphery of pulley **413**D and returns through tube **412**, seal **419**, and block **420** to terminate with a plurality of turns of cable **423** wound around the periphery of capstan anchor **426** before passing through a complementary hole in anchor **426**. Anchor **426** is disposed in a complementary groove in block **420** to prevent both rotation and translation of anchor **426**. The turns of cable **423** about anchor **426**

and drum **422** exploit the capstan effect to allow the complementary cable holes in anchor **426** and drum **422** to retain the respective ends of cable **423**. Seal **427** seals against the face of end cover **207** (see FIG. **14**) to prevent the ingress of external ambient air into block **320** through other than threaded port **212**B (see FIG. **14**). Tierods **418** surround tube **412** to mechanically join together end blocks **405** and **420**.

(40) Operation of piston assemblies **313** and **413** is analogous to the operation of piston assemblies of **114** and **115** in the remote drive of FIG. **2**. As piston assembly **313** moves directionally toward end cover **204**, the volume of air enclosed by block **305**, tube **312**, and piston assembly **313** is reduced, forcing the enclosed air through port **211**A. Simultaneously with air exiting through port **211**A, air flows through port **212**A, to fill the partial vacuum which is created as the volume of air enclosed by block **320**, tube **312**, and piston assembly **313** is increased. As piston assembly **313** moves directionally away from end cover **204**, the volume of air enclosed by block **305**, tube **312**, and piston assembly **313** is increased, with the friction exerted by seal **313**J acting against the inner wall of tube **312** causing inner piston **313**B to translate relative to outer piston **313**H. Such translation opens the annular passage between inner piston **313**B and outer piston **313**H, allowing external ambient air to pass into the volume enclosed by block **305**, tube **312**, and piston assembly **313**, so as to replenish any air that may have leaked from the pneumatic circuit formed by the pneumatic connection between the volume enclosed by block **305**, tube **312**, and piston assembly **313** and an externally connected pneumatically driven actuator (not shown). Analogously, as piston assembly **413** moves directionally toward end cover **204**, the volume of air enclosed by block **405**, tube **412**, and piston assembly **413** is reduced, forcing the enclosed air through port **211**B. Simultaneously with air exiting through port **211**B, air flows through port **212**B, to fill the partial vacuum which is created as the volume of air enclosed by block **420**, tube **412**, and piston assembly **413** is increased. As piston assembly **413** moves directionally away from end cover **204**, the volume of air enclosed by block **405**, tube **412**, and piston assembly **413** is increased, with the friction exerted by seal **413**J acting against the inner wall of tube **412** causing inner piston **413**B to translate relative to outer piston **413**H. Such translation opens the annular passage between inner piston **413**B and outer piston **413**H, allowing external ambient air to pass into the volume enclosed by block **405**, tube **412**, and piston assembly **413**, so as to replenish any air that may have leaked from the pneumatic circuit formed by the pneumatic connection between the volume enclosed by block **405**, tube **412**, and piston assembly **413** and an externally connected pneumatically driven actuator (not shown).

(41) Referring now to FIG. **18**, the cable system that moves piston assembles **313** and **413** is shown. Although not shown in FIG. **18**, it is understood that timing belt **206** will simultaneously rotate timing belt pulleys **301** and **401** and shafts **302** and **402** respectively, to which the pulleys **301**, **401** are attached, as the output shaft of motor **201** rotates (see also FIG. **14**). As motor **201** rotates shafts **302** and **402** clockwise (CW), additional turns of cable **308** will be wound around drum **307**, shortening the length of cable **308** spanning between pulley **309** and pulley **313**A and between pulley **313**A and anchor **314**. Reduction of cable **308** spans exert a force, equal to twice the tension in cable **308**, against pulley **313**A which causes piston assembly **313** to move in a direction towards pulley **309**. As shaft **302** and drum **307** rotate CW, drum **322** on the opposing end of shaft **302** simultaneously rotates CW, extending the length of cable **323** spanning between pulley **324** and pulley **313**D as piston assembly **313** moves away from pulley **324**. The outer diameters of drums **307** and **322** are chosen to be identical so that the additional amount of cable **308** wound upon drum **307** will always be equal to the amount of cable **323** released from drum **322**. In analogous manner, as shaft **402** rotates CW, additional turns of cable **423** will be wound around drum **422**, shortening the length of cable **423** spanning between pulley **424** and pulley **413**D and between pulley **413**D and anchor **426**. Reduction of the cable **423** spans exert a force, equal to twice the tension in cable **423**, against pulley **413**D which causes piston assembly **413** to move in a direction towards pulley **424**. As shaft **402** and drum **422** rotate CW, drum **407** on the opposing end

of shaft **202** simultaneously rotates CW, extending the length of cable **408** spanning between pulley **409** and pulley **413**A as piston assembly **413** moves away from pulley **409**. The outer diameters of drums **407** and **422** are chosen to be identical so that the additional amount of cable **423** wound upon drum **422** will always be equal to the amount of cable **408** released from drum **407**.

(42) It will be apparent to one skilled in the art that as motor **201** rotates shafts **302** and **402** counterclockwise (CCW), the motions of piston assemblies **313** and **413** will be reversed, with piston assembly **313** moving away from pulley **309** as piston assembly **413** moves toward pulley **409** under the action of the force applied by the two spans of cable **408** against pulley **413**A.

(43) Referring now to FIG. **19**, there is shown another embodiment of a drive system, namely, drive system **503**, which includes remote drive **200**, storage container **501** (which can be a reservoir or an accumulator but is assumed to be a reservoir herein), check valves **500**A, **500**B, and ports **211**A, **211**B, **212**A, **212**B, and port **502** (drive system **503** further includes an actuator (not shown) that is actuated by remote drive **200**). FIG. **19** shows the application of the remote drive **200** of FIG. **13** to provide a source of pressurized air. Ports **211**A and **211**B of remote drive **200** are connected through check-valves **500**A and **500**B respectively, such as those manufactured by the Check-All Company, to reservoir **501**. Storage container **501** is configured for storing a fluid medium, such as air, therein. Storage container **501** and check valves **500**A, **500**B are fluidically coupled with remote drive **200**. In operation, remote drive **200** is cycled to alternately produce compressed air from ports **211**A and **211**B. When the compressed air produced at port **211**A exceeds a selected pressure, check valve **500**A opens, allowing compressed air to flow from the remote drive **200** into reservoir **501**. Alternately, when the compressed air produced at port **211**B exceeds a selected pressure, check valve **500**B opens, allowing compressed air to flow from the remote drive **200** into reservoir **501**. Port **502**, attached to reservoir **501** allows the compressed air within the reservoir to be connected to an external pneumatic circuit (not shown). In this manner, cyclical operation of remote drive **200** results in a continuous flow of ambient air drawn into ports **212**A and **212**B exiting as compressed air from ports **211**A and **211**B, respectively, to maintain a supply of compressed air within reservoir **501**. This embodiment of the present invention thus replaces a pneumatic coupling between the remote drive **200** and the driven actuator with an accumulator or reservoir **501** in which a volume of air is stored after being compressed to a specified pressure by the action of the remote drive **200**.

(44) Referring now to FIG. **20**, there is shown another embodiment of a drive system, namely, drive system **603**, which includes remote drive **200**, storage container **601** (which can be a reservoir or an accumulator but is assumed to be a reservoir herein), check valves **600**A, **600**B, and ports **211**A, **211**B, **212**A, **212**B, and port **602**. FIG. **20** shows the application of the remote drive **200** of FIG. **13** to provide a source of rarified air (i.e. a partial vacuum). Ports **212**A and **212**B are connected through check-valves **600**A and **600**B respectively, such as those manufactured by the Check-All Company, to reservoir **601**. Storage container **601** is configured for storing a fluid medium, such as rarefied air, therein. Storage container **601** and check valves **600**A, **600**B are fluidically coupled with remote drive **200**. In operation, remote drive **200** is cycled to alternately produce rarified air from ports **212**A and **212**B. When the rarified air produced at port **212**A subceeds a selected level of partial vacuum, check valve **600**A opens, allowing air to flow from reservoir **601** into the remote drive **200**. Alternately, when the rarified air produced at port **212**B subceeds a selected level of partial vacuum, check valve **600**B opens, allowing air to flow from reservoir **601** into the remote drive **200**. Port **602**, attached to reservoir **601**, allows the partial vacuum within the reservoir **601** to be connected to an external pneumatic circuit (not shown). In this manner, cyclical operation of remote drive **200** results in a continuous flow of rarefied air drawn into ports **212**A and **212**B as ambient pressure air exits from ports **211**A and **211**B, respectively, to maintain a partial vacuum within reservoir **601**. This embodiment of the present invention thus replaces the pneumatic coupling between the remote drive **200** and the driven actuator with an accumulator or reservoir **601** in which a volume of rarified air is stored after being evacuated to a specified partial vacuum

by the action of the remote drive **200**. Accordingly, drive system **603**, which is operated to produce a partial vacuum, further includes an actuator (not shown) that is actuated by remote drive **200** and is coupled with port **602**; this actuator is configured for employing at least a partial vacuum and thus, according to an embodiment of the present invention, is a suction cup configured for grasping a workpiece when the suction cup is evacuated to exert a force against the workpiece to be grasped.

(45) Referring now to FIGS. **21-23**, there is shown another embodiment the drive system according to the present invention, namely, drive system **733**, drive system **733** including a remote drive **734** (schematically shown) according to the present invention (such as remote drives **1**, **100**, **200**), and an actuator according to an embodiment of the present invention, namely, actuator **700**. Actuator **700** is a pneumatically powered cable driven rotary actuator **700** according to the present invention. This embodiment of the actuator is an adaptation of remote drives **100** and/or **200** to form an actuator **700** wherein a source of compressed air is used to convert the linear motion of an internal piston (piston assembly **713**) into an externally accessible rotational motion provided by a drive shaft (drive shaft **701**) in order to drive a downstream device (see FIGS. **15-16** and **22**, wherein actuator **700** is structurally similar to cylinder assemblies **300**/**400**). Actuator can be a part of a drive system **733**, which further includes a supply of a fluid medium which is compressed or otherwise pressurized, such as compressed air. Actuator **700** includes, respectively, drive shaft **701**, radial bearing **702**, retaining ring **703**, end block **704**, radial seals **705**A, **705**B, drum **706**, cable **707**, cable pulley **708**, pivot pin **709**, cover **710**, O-ring seal **711**, cylinder tube **712**, piston assembly **713**, anchor **714**, seal **715**, retaining ring **716**. Radial bearing **717**, tierods **718**, O-ring seal **719**, end block **720**, radial seal **721**, cable drum **722**, cable **723**, cable pulley **724**, pivot pin **725**, anchor **726**, seal **727**, end cover **728**, fasteners **729**, **730**, and ports **731**, **732**. Piston assembly **713** includes cable pulley **713**A, piston **713**B, pivot pin **713**C, cable pulley **713**D, pivot pin **713**E, and seal **713**F. Piston assembly **713**, cables **707**, **723**, and drive shaft **701** are coupled with one another, piston assembly **713** and cables **707**, **723** together being configured for causing drive shaft **701** to rotate. In general, to move shaft **701** CW, air moves into cylinder tube **712** by way of port **731**, which moves piston assembly **713** to the right in FIG. **22**, causing cable **707** to unwind and drum **706** to move CW, which moves shaft **701** CW, which also moves drum **722** CW, causing cable **723** to wrap around drum **722**. To move shaft **701** CCW, air moves into cylinder tube **712** by way of port **732**, which pushes piston assembly **713** to the left in FIG. **22**, causing cable **723** to unwind and drum **722** to move CCW, which moves shaft **701** CCW, which also moves drum **706** CCW, causing cable **707** to wrap around drum **706**. One end of drive shaft **701** passes though radial bearing **702**, with bearing **702** retained by retaining ring **703** within a complementary bore in end block **704**. The end of shaft **701** then passes through radial seal **705**A, disposed in a complementary bore in block **704** so as to prevent air from passing between the shaft and block, and into cable drum **706**, with setscrews (not shown) securing drum **706** onto the end of shaft **701**. Shaft **701** then passes through radial seal **705**B, disposed in a complementary bore in cover **710** so as to prevent air from passing between shaft **701** and cover **710** and out of cover **710** so as to allow a mechanical connection of shaft **701** to an external component (not shown) to be rotated. One end of cable **707** passes through a hole in drum **706**, with a plurality of turns of cable **707** wound around the periphery of drum **706**. Cable **707** may take the form of steel wire rope or be constructed from suitable polymer threads woven or braided together. Cable **707** is directed from the windings about drum **706** to wrap around the outer diameter of cable pulley **708**, which is retained in a complementary slot in end block **704** by, and is free to pivot about, pivot pin **709** which is disposed in a complementary hole in the protruding boss of block **704**. Cable **707** passes through block **704**, O-ring seal **711** and cylinder tube **712** to engage cable pulley **713**A, which is retained in a complementary slot in piston **713**B by, and is free to pivot about, pivot pin **713**C which is disposed in a complementary hole in piston **713**B (see FIG. **23**). O-ring seal **711** seals the end of tube **712** against the face of end block **704** to prevent the egress of compressed air from the end of tube **712**. Cable **707** continues around the periphery of pulley **713**A and returns through tube **712**, seal **711**,

and block **704** to terminate with a plurality of turns of cable **707** wound around the periphery of capstan anchor **714** before passing through a complementary hole in anchor **714**. Anchor **714** is disposed in a complementary groove in block **704** to prevent both rotation and translation of anchor **714**. The turns of cable **707** about the body of anchor **714** exploit the so-called capstan effect, wherein any tension applied to cable **707** is progressively reduced by friction between the surface of cable **707** and the surface of anchor **714** as additional turns of cable **707** are added. This reduces the tension applied to cable **707** by the action of pulley **713**A acting upon cable **707** to a level sufficient to retain the end of cable **707** within the hole in anchor **714** through which cable **707** passes. In an analogous manner, the turns of cable **707** wound about cable drum **706** also exploit the capstan effect to retain the opposing end of cable **707** within the complementary cable hole in the drum **706**. Seal **715** seals against the face of end cover **710** to prevent the egress of compressed air from the block **704** through other than threaded port **731**. Seal **713**F seals the periphery of piston **713**B against the inner diameter of cylinder tube **712**. Further, threaded fasteners **729**, **730** attach end covers **728**, **710** to end blocks **720**, **704**, respectively.

(46) The opposing end of drive shaft **701** passes though radial bearing **717**, with the bearing retained by retaining ring **716** within a complementary bore in end block **720**. The end of shaft **701** then passes through radial seal **721**, disposed in a complementary bore in block **720** so as to prevent air from passing between shaft **701** and block **720**, and into cable drum **722**, with setscrews (not shown) securing drum **722** onto the end of shaft **701**. One end of cable **723** passes through a hole in drum **722**, with a plurality of turns of cable **723** wound around the periphery of drum **722**. Cable **723** may take the form of steel wire rope or be constructed from suitable polymer threads woven or braided together. Cable **723** is directed from the windings about drum **722** to wrap around the outer diameter of cable pulley **724**, which is retained in a complementary slot in end block **720** by, and is free to pivot about, pivot pin **725** which is disposed in a complementary hole in the protruding boss of block **720**. Cable **723** passes through block **720**, O-ring seal **719**, and cylinder tube **712** to engage cable pulley **713**D (FIG. **23**), which is retained in a complementary slot in inner piston **713**B by, and is free to pivot about, pivot pin **713**E which is disposed in a complementary hole in piston **713**B. O-ring seal **719** seals the opposing end of tube **712** against the face of end block **720** to prevent the egress of compressed air from the end of tube **712**. Cable **723** continues around the periphery of pulley **713**D and returns through tube **712**, seal **719**, and block **720** to terminate with a plurality of turns of cable **723** wound around the periphery of capstan anchor **726** before passing through a complementary hole in anchor **726**. Anchor **726** is disposed in a complementary groove in block **720** to prevent both rotation and translation of anchor **726**. The turns of cable **723** about anchor **726** and drum **722** exploit the capstan effect to allow the complementary cable holes in anchor **726** and drum **722** to retain the respective ends of cable **723**. Seal **727** seals against the face of end cover **728** to prevent the egress of external compressed air from block **720** through other than threaded port **732**. Tierods **718** surround tube **712** to mechanically join together end blocks **704** and **720**.

(47) During operation of the rotary actuator **700**, compressed air is directed into the volume formed by end block **704**, tube **712**, and movable piston **713** though the threaded port hole **731** in cover **710**. The compressed air, acting on the face of piston **713** closest to block **704**, creates a force acting to push piston **713** away from block **704**. As piston **713** moves away from block **704**, a proportional force is applied to cable **707** to rotate drum **706** clockwise (CW) removing turns of cable **707** from drum **706** as drum **706** rotates. Drum **706**, which is mechanically attached to shaft **701**, correspondingly rotates shaft **701** CW. As compressed air is applied to port **731**, the air within the volume formed by piston **713**, tube **712**, and opposing end block **720** is simultaneously exhausted through threaded port **732** in opposing cover **728**. The rotation of shaft **701** induces a corresponding CW rotation of opposing drum **722**, which is mechanically attached to shaft **701**. The CW rotation of drum **722** winds turns of cable **723** upon drum **722** at the same rate as the lengths of cable **723** spanning between pulley **713**D within piston **713** and anchor **726** and between

pulley **713**D and drum **722** are shortened by the translation of piston **713**.
(48) To rotate shaft **701** in a counterclockwise (CCW) direction, compressed air is directed into the volume formed by end block **720**, tube **712**, and movable piston **713** though the threaded port hole **732** in cover **728**. The compressed air, acting on the face of piston **713** closest to block **720**, creates a force acting to push piston **713** away from block **720**. As piston **713** moves away from block **720**, a proportional force is applied to cable **723** to rotate drum **722** counterclockwise (CCW) removing turns of cable **723** from drum **722** as drum **722** rotates. Drum **722**, which is mechanically attached to shaft **701**, correspondingly rotates shaft **701** CCW. As compressed air is applied to port **732**, the air within the volume formed by piston **713**, tube **712**, and end block **704** is simultaneously exhausted through threaded port **731** in opposing cover **710**. The rotation of shaft **701** induces a corresponding CCW rotation of drum **706**, which is mechanically attached to shaft **701**. The CCW rotation of drum **706** winds turns of cable **707** upon drum **706** at the same rate as the lengths of cable **707** spanning between pulley **713**A within piston **713** and anchor **714** and between pulley **713**A and drum **706** are shortened by the translation of piston **713**.
(49) Referring now to FIGS. **24-28**, there is shown another embodiment the drive system according to the present invention, namely, drive system **853**, drive system **853** including another embodiment of the remote drive according to the present invention, namely, remote drive **800**, and an actuator **854**. Remote drive **800** includes the embodiment of the remote drive **200** (see FIGS. **13** and **14**) with additional ways to guide cables **308**, **323**, **408**, and **423** (see FIGS. **15** and **16**) as they are wound and unwound about drums **307**, **322**, **407**, and **422**, respectively. Remote drive **800** is substantially similar in construction to remote drive **200**, with the exception for instance of covers **804** and **807** substituting for covers **204** and **207**, respectively, and cylinder assemblies **900** and **1000** substituting for cylinder assemblies **300** and **400**, respectively. Remote drive **800** additionally incorporates dowel pins **852** and cable guide assemblies **850**/**851** (see also FIG. **26**). Thus, remote drive **800** includes a motor (shown but unnumbered) and a belt (shown but unnumbered) which is driven by the motor and which drives shafts **902** and **1002**. Further, remote drive **800** includes covers **804** (including cavities **804**A, **804**B), **807**, cable guide assemblies **850**, **851**, dowl pin **852**, cylinder assemblies **900**, **1000**, drums **907**, **1007**, cables **908**, **1008**, drums **922**, **1022**, cables **923**, **1023**, and piston assemblies **913**, **1013**. Cable guide assembly **850** includes guide body **850**A, nut **850**B, internal thread **850**C, dowel pin **850**D, pulley **850**E, busing **850**F, and set screws **850**G. Cable guide assembly **851** is substantially similar to cable guide assembly **850** (unless otherwise shown or stated herein) and thus also includes nut **851**B and pulley **851**E. Shafts **902**, **1002** include, respectively, threaded ends **902**A, **1002**A, **902**B, **1002**B. Further, remote drive **800** includes a cable and pulley system **855**, piston assemblies **913**, **1013** being coupled with cable and pulley system **855**. Cable and pulley system **855** includes cables **908**, **1008**, **922**, **1022** and cable guide assemblies **850**, **851** (structure overlapping between cable and pulley system **855** and piston assemblies **913**, **1013** can be regarded as being a part of cable and pulley system **215** or piston assemblies **313**, **413**). Piston assemblies **913**, **1013** are substantially similar to piston assemblies **313**, **413** (unless otherwise shown or stated herein), and thus piston assemblies **913**, **1013** also include inner pistons and outer pistons which are configured for sliding relative to one another and thereby for selectively opening and closing a working medium passage therebetween (see FIGS. **9-12**, for, the structure, function, and operation of working medium passage **129** of piston assemblies **313**, **413** are substantially similar to those of working medium passage **129** of piston assemblies **114**, **115**). In general, the motor of remote drive **800** moves the belt, which rotates drive shafts **902**, **1002** and drums **907**/**1007**, **922**/**1022**, causing spans of cables **908**/**1008** and **923**/**1023** to lengthen or shorten, which causes piston assemblies **913**, **1013** (with slots and pins substantially similar to slots **313**G, **413**G and pins **313**C, **413**C) to translate within respective tubes, which causes air to flow in or out of ports (which are substantially similar to ports **211**A/**211**B, **212**A/**212**B), so as to move a working device thereby, such as an actuator similar to actuator **2**. This embodiment of the remote drive thus provides an adaptation of remote drive **200** to add a way to guide the cables **908**, **923**, **1008**, **1023**

of the cable and pulley system as the cables **908**, **923**, **1008**, **1023** are wound respectively around drums **907**, **922**, **1007**, **1022** to convert the rotational motion from the electric motor into linear motion of the piston assemblies **913**, **1013** fluidically coupled to a pneumatically driven actuator.
(50) Referring now to FIGS. **26** and **27**, dowel pins **852** are press-fit into complementary bores (not shown) in cavities **804**A/**804**B. Cable guide assemblies **850** and **851** are slidably deposed into complementary cavities **804**A and **804**B, respectively, around the protruding portions of dowel pins **852**.
(51) Referring now to FIGS. **27** and **28**, bushings **850**F, which are press-fit into a complementary bore in guide body **850**A, slidably engage dowel pin **852** to prevent the cable guide assembly **850** from pitching about the longitudinal axis of the dowel pin **852**, while allowing the cable guide assembly **850** to freely traverse the length of the dowel pin **852**. The slidable engagement of guide body **850**A within the confines of cavity **804**A (see also FIG. **26**) prevents cable guide assembly **850** from rotating about dowel pin **852**, while allowing the cable guide assembly **850** to freely traverse the length of cavity **804**A. In a similar manner, bushings (not shown), press-fit into a complementary bore in guide body **851** and slidably engage dowel pin **852** to prevent the cable guide assembly **851** from pitching about the longitudinal axis of the dowel pin **852**, while allowing the cable guide assembly **851** to freely traverse the length of the dowel pin **852**. The slidable engagement of cable guide assembly **851** within the confines of cavity **804**B (see also FIG. **26**) prevents the cable guide assembly **851** from rotating about dowel pin **852**, while allowing the cable guide assembly **851** to freely traverse the length of cavity **804**B. It is understood that cover **807** also contains press-fit dowel pins **852** and cavities (not shown) of identical geometry to those of cavities **804**A and **804**B, which slidably engage guide assemblies **850** and **851** to prevent pitching of the guide assemblies **850**, **851** about the longitudinal axes of the dowel pins **852** and restrict rotation of guide assemblies **850**, **851** about the dowel pins **852**.
(52) Dowel pin **850**D is press-fit into a complementary hole in guide body **850**A. A complementary hole in pulley **850**E engages dowel pin **850**D in a manner that allows pulley **850**E to freely rotate about the dowel pin **850**D. Nut **850**B, containing internal thread **850**C, slip-fits into a complementary bore in guide body **850**A and is retained after insertion into the bore by setscrew **850**G which passes through a threaded complementary hole in guide body **850**A. This arrangement between nut **850**B and guide body **850**A allows the nut **850**B to be threaded onto complementary thread **902**B present on the end of drive shaft **902** (see also FIG. **26**). Once the nut **850**B has been installed onto threaded portion **902**B of shaft **902** to the desired location, guide assembly **850** is placed onto the nut **850**B, and setscrew **850**G is tightened to retain the guide assembly **850** onto the nut **850**B. In an analogous manner, a second nut **850**B is threaded onto complementary thread **902**A (see FIG. **28**) placed on the end of shaft **902**, opposite thread **902**B, and a second guide assembly **850** is similarly retained on the nut **850**B.
(53) Referring now to FIG. **28**, cable guide pulley **850**E rests against the surface of cables **908**/**923** with the position of the cable **908**, **923** relative to drums **907**/**922** determined by the location of guide assembly **850**. Thread **850**C (see FIG. **27**) is selected to be a right-hand thread on both of the nuts **850**B which engage the threaded ends **902**A/**902**B of shaft **902**. The pitch of thread **850**C is selected to be equal to, or slightly greater than, the diameters of cables **908**/**1008** and **923**/**1023**. As shaft **902** is rotated CW during operation of remote drive **800**, additional cable **908** is wound onto drum **907**. Simultaneously, the action of thread **805**C against thread **902**A causes guide assembly **850** to move away from drum **907** at a rate governed by the pitch of the thread, so that as drum **907** rotates one turn, cable **908** is allowed to advance approximately one cable diameter as it winds upon drum **907**. This action precludes any tendency of the turns of cable **908** to lay over top of one another as additional cable **908** is wound onto the drum **907**. In a similar manner, CW rotation of shaft **902** also causes guide assembly **850** on the opposite end of shaft **902** to move toward drum **922**, allowing for a controlled unwinding of cable **923** from drum **922**.
(54) It will be apparent to one skilled in the art that CCW rotation of shaft **902** produces an

opposite effect, with the winding of cable **923** onto drum **922** controlled by the progressive translation of the corresponding guide assembly **850** away from drum **922** and the unwinding of cable **908** from drum **907** controlled by the progressive translation of the corresponding guide assembly **850** toward drum **907**.

(55) Cable guide assemblies **851** are identical in construction to guide assemblies **850**, with the exception for instance that a left-handed thread is chosen for the thread (not shown) in nut **851**B and complementary threads **1002**A/**1002**B on the ends of shaft **1002**. In an analogous manner to the operation of shaft **902**, guide assembles **850**, drums **907** and **922** and cables **908** and **923**, CW rotation of shaft **1002** causes additional turns of cable **1023** to be wound about drum **1022**, while guided and controlled by the action of pulley **851**E acting against cable **1023**, while cable **1008** is simultaneously unwound from drum **1007**, while guided and controlled by the action of pulley **851**E acting against cable **1008**. CCW rotation of shaft **1002** causes additional turns of cable **1008** to be wound about drum **1007**, while guided and controlled by the action of pulley **851**E acting against cable **1008**, while cable **1023** is simultaneously unwound from drum **1022**, while guided and controlled by the action of pulley **851**E acting against cable **1023**.

(56) Referring now to FIG. **29**, there is shown a flow diagram showing a method **1160** of using a drive system **18**, **126**, **213**, **503**, **603**, **733**, **853**, the method **1160** including the steps of: providing **1161** that the drive system **18**, **126**, **213**, **503**, **603**, **733**, **853** includes a remote drive **1**, **100**, **200**, **734**, **800** and an actuator **2**, **127**, **214**, **700**, **854**, the remote drive **1**, **100**, **200**, **734**, **800** being configured for being driven by an electrical motor **14**, **101**, **201**, the actuator **2**, **127**, **214**, **700**, **854** being spaced apart from and fluidically coupled with the remote drive **1**, **100**, **200**, **734**, **800**; and fluidically powering **1162** the actuator **2**, **127**, **214**, **700**, **854** by the remote drive **1**, **100**, **200**, **734**, **800**. The remote drive **1**, **100** can include a rack and pinion system **19**, **128** and a plurality of piston assemblies **20**A, **20**B, **114**, **115** coupled therewith, the plurality of piston assemblies **20**A, **20**B, **114**, **115** each including an inner piston **114**B, **115**B and an outer piston **114**D, **115**D which are configured for sliding relative to one another and thereby for selectively opening and closing a working medium passage **129** therebetween. The remote drive **200** can include a cable and pulley system **215**, **855** and a plurality of piston assemblies **313**, **413** coupled therewith, the plurality of piston assemblies **313**, **413** each including an inner piston **313**B, **413**B and an outer piston **313**H, **413**H which are configured for sliding relative to one another and thereby for selectively opening and closing a working medium passage **129** therebetween. The cable and pulley system **855** can further include a plurality of cable guide assemblies **850**, **851**. The drive system **213**, **503**, **603**, **853** can further include a storage container **501**, **601** and a plurality of check valves **500**A, **500**B, **600**A, **600**B, the storage container **501**, **601** and the plurality of check valves **500**A, **500**B, **600**A, **600**B being fluidically coupled with the remote drive **200**, the storage container **501**, **601** being configured for storing a fluid medium therein, the fluid medium being air or rarified air. The actuator **700** can include a piston assembly **713**, a plurality cables **707**, **723**, and a drive shaft **701** coupled with one another, the piston assembly **713** and the plurality of cables **707**, **723** together being configured for causing the drive shaft **701** to rotate.

(57) While this invention has been described with respect to at least one embodiment, the present invention can be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains and which fall within the limits of the appended claims.

Claims

1. A drive system, comprising: a remote drive configured for being driven by an electrical motor, wherein the remote drive includes a cable and pulley system and a plurality of piston assemblies

coupled therewith, the plurality of piston assemblies each including an inner piston and an outer piston which are configured for sliding relative to one another and thereby for selectively opening and closing a working medium passage therebetween; and an actuator spaced apart from and fluidically coupled with the remote drive and configured for being fluidically powered by the remote drive.

2. The drive system according to claim 1, wherein the cable and pulley system further includes a plurality of cable guide assemblies.

3. The drive system according to claim 1, further including a storage container and a plurality of check valves, the storage container and the plurality of check valves being fluidically coupled with the remote drive, the storage container being configured for storing a fluid medium therein, the fluid medium being air or rarified air.

4. A drive system, comprising: a remote drive configured for being driven by an electrical motor; and an actuator spaced apart from and fluidically coupled with the remote drive and configured for being fluidically powered by the remote drive, wherein the actuator includes a piston assembly, a plurality of cables, and a drive shaft coupled with one another, the piston assembly and the plurality of cables together being configured for causing the drive shaft to rotate.

5. A drive of a drive system, the drive comprising: the drive, which is a remote drive that is configured for being driven by an electrical motor, is configured for being spaced apart and fluidically coupled with an actuator of the drive system, and is configured for fluidically powering the actuator, wherein the remote drive includes a cable and pulley system and a plurality of piston assemblies coupled therewith, the plurality of piston assemblies each including an inner piston and an outer piston which are configured for sliding relative to one another and thereby for selectively opening and closing a working medium passage therebetween.

6. The drive according to claim 5, wherein the cable and pulley system further includes a plurality of cable guide assemblies.

7. The drive according to claim 5, wherein the remote drive is configured such that the drive system further includes a storage container and a plurality of check valves such that the storage container and the plurality of check valves are fluidically coupled with the remote drive and the storage container is configured for storing a fluid medium therein, which is air or rarified air.

8. A drive of a drive system, the drive comprising: the drive, which is a remote drive that is configured for being driven by an electrical motor, is configured for being spaced apart and fluidically coupled with an actuator of the drive system, and is configured for fluidically powering the actuator, wherein the remote drive is configured such that the actuator includes a piston assembly, a plurality of cables, and a drive shaft coupled with one another, such that the piston assembly and the plurality of cables together are configured for causing the drive shaft to rotate.

9. A method of using a drive system, the method comprising the steps of: providing that the drive system includes a remote drive and an actuator, the remote drive being configured for being driven by an electrical motor, the actuator being spaced apart from and fluidically coupled with the remote drive, wherein the remote drive includes a cable and pulley system and a plurality of piston assemblies coupled therewith, the plurality of piston assemblies each including an inner piston and an outer piston which are configured for sliding relative to one another and thereby for selectively opening and closing a working medium passage therebetween; and fluidically powering the actuator by the remote drive.

10. The method according to claim 9, wherein the cable and pulley system further includes a plurality of cable guide assemblies.

11. The method according to claim 9, wherein the drive system further includes a storage container and a plurality of check valves, the storage container and the plurality of check valves being fluidically coupled with the remote drive, the storage container being configured for storing a fluid medium therein, the fluid medium being air or rarified air.

12. A method of using a drive system, the method comprising the steps of: providing that the drive

system includes a remote drive and an actuator, the remote drive being configured for being driven by an electrical motor, the actuator being spaced apart from and fluidically coupled with the remote drive, wherein the actuator includes a piston assembly, a plurality of cables, and a drive shaft coupled with one another, the piston assembly and the plurality of cables together being configured for causing the drive shaft to rotate; and fluidically powering the actuator by the remote drive.